

There are total 50 questions in this Test Booklet. Each question is having four alternative responses, of which one response is completely correct. Record the completely correct response on the OMR sheet by darkening the corresponding circle with Blue/Black ball point pen only.

1. Which seas have higher salinity
- (A) Arctic Ocean, Persian Gulf, Bering Sea
- (B) Red Sea, North Sea, Mediterranean Sea
- (C) Red Sea, Persian Gulf, Mediterranean Sea
- (D) Red Sea, Persian Gulf, Bering Sea
2. What is the primary difference between covariance and correlation?
- (A) Covariance measures the strength of a relationship, while correlation measures the direction
- (B) Covariance is standardized, while correlation is not
- (C) Correlation is standardized, while covariance is not
- (D) Covariance is always positive, while correlation can be positive or negative
3. The reason/s of Northeast region of India is seen as an emerging hub of palm oil plantations is/are
- (a) Congenial physiography and climate
- (b) Initiative of government of India
- (c) Have abundant open land for palm plantations
- (d) A ready market for palm oil in the region
- (A) Only statement (a) is correct (B) Only statement (b) is correct
- (C) Statements (a) and (b) are correct (D) All statements are correct
4. Which one of the following electromagnetic wavelengths are the longest?
- (A) Radio waves (B) Gamma rays
- (C) Infrared (D) Visible
5. The concept of Lebensraum became popular in the discipline of political geography after the publication of:
- (A) Making Political Geography (B) Geographical Pivot of History
- (C) Cosmography (D) Politische Geographic

✓ 6. The following statements are about general pattern of ocean currents

- i) Equatorial currents move west to east ✗
- ii) In the lower latitudes warm currents are found on the eastern shores of the continents
- iii) In the higher latitudes warm currents flow along the western shores ✓
- iv) Upwelling is most common along east coast of continents.

Which of the above statements are correct

- ~~+~~(A) ~~(i) and (ii)~~ ✗
- (B) (ii) and (iii)
- (C) (iii) and (iv)
- ~~✗~~(D) ~~(i) and (iv)~~

7. Which of the following statements is true about correlation and causation?

- (A) Correlation does not imply causation
- (B) Correlation always implies causation
- (C) Correlation sometimes implies causation
- (D) Correlation and causation are the same

✓ 8. Malthusian theory in the late the 18th said that of population growth at some point of time would outstrip food supply and other resources; the point what would be called Malthusian catastrophe.

Consider the following statements-

- a) Malthus failed to read the rapid stride of scientific and technological progress of his times and the future. ✓
- b) Finite, non-renewability and non-reproducibility of certain natural resources indicate that technology has a limit, and Malthus was not wrong.
- ✓ c) Crises of present times - poverty, famine, war etc., indicate Malthus was right.
- d) Malthus provided a useful construct to realise the limit of our world.

- (A) Only (a) is correct
- (B) (b) and (c) are correct
- (C) (b), (c) and (d) are correct
- (D) All are correct

9. Rayleigh scattering occurs when EMR interacts with Atmospheric particles are-

- (A) Equal to wavelength
- (B) Larger than wavelength
- (C) Smaller than wavelength
- (D) Question does not provide sufficient data or is vague

10. The Organic Theory of state was propounded by:

- (A) Ratzel (B) Mackinder
(C) Isaiah Bowman (D) Haushoffer

11. Which of the following processes of soil formation causes translocation of soil materials

- (A) Podsolisation (B) Calcification
(C) Weathering (D) Humification

12. What does the term "residual" mean in regression analysis?

- (A) The predicted value of the dependent variable
(B) The difference between the observed and predicted values of the dependent variable
(C) The slope of the regression line
(D) The intercept of the regression line

13. Consider a country at the 4th stage of Demographic Transition to expand its emerging economy immediately would like to resort to

- (a) Encourage immigration of skilled workforce
(b) Its population policy would provide liberal childcare leaves to its existing workers

- (A) Approach a will be rational (B) Approach b will be rational
(C) Both approaches will be rational (D) Both approaches are irrational

14. Aurora Australis pertaining to ionosphere are related to which hemisphere/region?

- (A) Northern Hemisphere (B) Greenland
(C) Southern Hemisphere (D) Canada

15. Assertion (A): Biomes are terrestrial ecosystems with specific climate, vegetation and animal life.

Reason (R): Temperature and precipitation determine the soil and vegetation of an area

- (A) Both (A) and (R) are true and R is the correct explanation of (A)
(B) Both (A) and (R) are true but (R) is the not correct explanation of (A)
(C) (A) is true but (R) is false
(D) Both (A) and (R) are false

16. Which of the following statements is true about ordinal scales?

- (A) They indicate relative ranks and order of the categories
- (B) They provide information about the exact differences between categories
- (C) They allow for the calculation of mean and standard deviation
- (D) They do not allow for meaningful comparisons of magnitude between categories

17. In India during 2001 to 2011 the number of Census Town showed rapid growth (increased by a total of 2500 in number) compared to the statutory towns (increased by around 250 in number).

Rapid growth of census towns was because of

- (a) Formation of municipal board/committee and notified town area committee, etc.
- (b) Rapid growth of population leading to cross the minimum requirement of 5000 inhabitants to qualify as a town
- (c) The town area become densely populated with more than 400 persons per square km
- (d) Three fourth of the male workforce found full time jobs in beyond agriculture activities

- (A) Statements (a), (b) and (c) are correct
- (B) Statements (b), (c) and (d) are correct
- (C) Statements (a), (c) and (d) are correct
- (D) Only (d) is correct

18. Which one of the following in the correct sequence in the development of the German School of thought?

- (A) Ratzel-Peschel-Hettner-Troll
- (B) Peschel-Ratzel-Hettner-Troll
- (C) Ratzel-Troll-Peschel-Hettner
- (D) Troll-Peschel-Hettner-Troll

19. Assertion (A): In stratosphere and thermosphere temperature increases with height.
Reason (R) : Absorption of sun's ultra-violet rays causes the increase in temperature.

Choose the most appropriate answer from the options given below

- (A) Both (A) and (R) are true but (R) is the not correct explanation of (A)
- (B) Both (A) and (R) are true and R is the correct explanation of (A)
- (C) (A) is true but (R) is false
- (D) Both (A) and (R) are false

20. Which map projection is known for preserving angles and shapes but distorting area, particularly near the poles?
- (A) Simple Cylindrical projection
 - (B) Lambert conformal conic projection
 - (C) Equal-area projection
 - (D) Mercator projection
21. Consider the following mode of cultivations
- (a) Jhum or shifting cultivation
 - (b) Alder based shifting cultivation
 - (c) Terrace cultivation
 - (d) Wet paddy cultivation
- (A) Only mode (a) is found in hill states of Northeast India
- (B) Mode (a) and (b) are found in hill states of Northeast India
- (C) Mode (a), (b) and (c) are found in hill states of Northeast India
- (D) All mode found in hill states of Northeast India.
22. Assertion (A): Tropical cyclones rarely form near the equator, although equator is a zone of wind convergence.
- Reason (R): The weak or no Coriolis force inhibit the development of a vortex.
- Choose the most appropriate answer from the options given below
- (A) (A) is true but (R) is false
 - (B) Both (A) and (R) are false
 - (C) Both (A) and (R) are true but (R) is the not correct explanation of (A)
 - (D) Both (A) and (R) are true and R is the correct explanation of (A)
23. If the bearing from point A to point B is 135 degrees, what is the bearing from point B to point A?
- (A) 45 degrees
 - (B) 225 degrees
 - (C) 315 degrees
 - (D) 135 degrees

24. The often reported rising intensities of the floods and damages in Assam said to have two explanations.

- a. The water flow has become sudden and high.
- b. People have established settlements in low lying areas.

- (A) Only (a) is right
- (B) Only (b) is right
- (C) Both (a) and (b) are true and inclusive to explain the causes of rising intensity.
- (D) (a) and (b) not inclusive statements to explain intensity of floods

25. Which of the following statements is true regarding geocentric latitude?

- (A) It is always greater than geodetic latitude
- (B) It is always less than geodetic latitude
- (C) It is equal to geodetic latitude at the equator
- (D) It is equal to geodetic latitude at the poles

✓ 26. **Assertion (A):** Sensors used in remote sensing can be classified based on the electromagnetic spectrum they detect.

Reason (R): Sensors sensitive to the visible spectrum are ideal for night-time imaging.

Options:

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is not the correct explanation of A
- (C) A is true but R is false
- (D) A is false but R is true

27. What is the angle between magnetic north and true north called?

- (A) Inclination
- (B) Variation
- (C) Declination
- (D) Deviation

28. Match the following-

- a. Spectral Resolution
- b. Spatial Resolution
- c. Temporal Resolution
- d. Radiometric Resolution

- I. Ability to define fine time intervals
- II. Ability to define fine intensity intervals
- III. Ability to define fine spatial intervals
- IV. Ability to define fine wavelength intervals

Options:

- (A) a(II), b(III), c(I), d(IV)
- (B) a(IV), b(III), c(I), d(II)
- (C) a(III), b(II), c(II), d(I)
- (D) a(II), b(II), c(III), d(IV)

29. How does the geoid relate to sea level?

- (A) The geoid is a fixed surface above sea level
- (B) The geoid has no relation to the sea level
- (C) The geoid represents the earth's surface when averaged to sea level
- (D) The geoid represents the sea surface when averaged to the earth's elevation

30. Which of the following is the correct sequence in the development of Geographic thought

- (A) Greek, Roman Arabs, Dark age
- (B) Greeks, Roman, Dark age, Arabs
- (C) Roman, Greeks, Arabs, Dark age
- (D) Greek, Dark Age, Roman, Arab

31. Which of the following statements is not a concept of Uniformitarianism

- (A) Natural laws are constant in time and space
- (B) Everything in nature can be explained scientifically
- (C) The earth has been shaped during a short period of time
- (D) There is uniformity of process and state

32. One of the demerits of quantitative methods is we may lose sight on an individual observation and miss certain specific components of the observation due to the use of group average or generalisation. This demerit can be appropriately addressed by

- (a) Addressing sampling error
- (b) Complementing with qualitative approaches of study
- (c) Adopting alternate statistical measures

- (A) Only through (a)
- (B) Only through (b)
- (C) Only through (c)
- (D) All the three approaches will be required

33. In the context of a normal distribution, what does the empirical rule state?
- (A) About 68% of the data falls within two standard deviations of the mean
 - (B) About 95% of the data falls within one standard deviation of the mean
 - (C) About 68% of the data falls within one standard deviation of the mean
 - (D) About 99.7% of the data falls within one standard deviation of the mean
34. Which of the following statements about digital image processing are correct?
- a. It involves the manipulation of digital images to enhance their quality and extract information.
 - b. It can be used to correct distortions caused by sensor irregularities.
 - c. It does not include the classification of image pixels.
 - d. It includes the transformation of image data into different representations.
- Options:
- (A) (a) and (c)
 - (B) (b) and (d)
 - (C) (a), (b), and (d)
 - (D) (b) and (c)
35. ✓ A scholar of which origin coined the term "permafrost" to explain the frozen characteristics of the Siberian soils?
- (A) American
 - (B) British
 - (C) French
 - (D) German
36. ✓ In 'Normal Cycle of Erosion' by Davis the relative relief is maximum during
- (A) Youth Stage
 - (B) Early Mature Stage
 - (C) Late Mature Stage
 - (D) Old stage
37. If the standard deviation of a normal distribution is 4 and the mean is 20, what is the z-score of a value of 28?
- (A) 1
 - (B) 2
 - (C) -1
 - (D) -2

38. To study a specific case, place, or a phenomenon the approach in geography would be Ideographic

- I. Descriptive and subjective experiences in explanations cannot be empirically tested
- II. Hence ideographic approach cannot make valid contributions to the field of knowledge

- (A) Statement I is wrong
- (B) Statement II is wrong
- (C) Statements I and II both are wrong
- (D) Statements I and II both are right

39. What is the term used to describe the ability of a sensor to define fine wavelength intervals?

- (A) Radiometric Resolution
- (B) Spatial Resolution
- (C) Spectral Resolution
- (D) Temporal Resolution

40. Identify the scholar who laid the foundation of the dichotomy of Systematic vs Regional Geography.

- (A) Cook
- (B) Varenus
- (C) Kant
- (D) Darwin

41. Lithosphere consists of

- (A) Continental crust
- (B) Oceanic crust
- (C) Both continental and oceanic crust
- (D) Continental crust, Oceanic crust and upper solid layer of mantle

42. Which measure of central tendency can be used for both numerical and categorical data?

- (A) Mode
- (B) Median
- (C) Mean
- (D) All of the above

43. Which of the following statements are unlikely

- (a) Population of India shall get doubled in every 35-40 years at annual growth rate of about 2 percent
- (b) Food grains yield, and production shall keep pace with population growth rate at least for next few decades
- (c) Per rural households' average availability of cultivable land will remain intact
- (d) Research and Technology will help to ensure food gain availability for infinite time

- (A) All are unlikely
- (B) (a) and (b)
- (C) (b) and (c)
- (D) (c) and (d)

- ✓ 44. Spatial data is stored in the computer by using
- (A) Plotter, scanner, keyboard
 - (B) Scanner, digitizer, keyboard
 - (C) Scanner and plotter
 - (D) All of the above

✓ 45. Match List I with List II:

List I	List II
a. Von Thunen	I. Central Place Theory
b. August Losch	II. Agricultural Location
c. Doi	III. Theory of Industrial Location
d. Christaller	IV. Crop combination

Choose the correct answer from the options given below:

- (A) a-II, b-IV, c-III, d-I
 - (B) a-II, b-I, c-IV, d-III
 - ✓ (C) a-II, b-III, c-IV, d-I
 - (D) a-I, b-II, c-III, d-IV
- ✓ 46. Which of the following statement on isostasy and eustasy is not correct
- (A) Due to melting of continental ice sheets there may be local rise of land areas ✓
 - (B) Both isostatic change and eustatic change are worldwide
 - (C) Sea level change occurs due to both isostatic and eustatic change
 - (D) Change in volume of ocean water causes eustatic rise or fall of sea level ✓
47. Which measure of dispersion is resistant to outliers?
- (A) Range
 - (B) Variance
 - (C) Standard Deviation
 - (D) Interquartile Range

48. The state governments of the Northeast region face serious challenges to accommodate their rising population pressure. Which of the following measures would help the state -

- ☒ (a) Raising efficiency and market reach of the existing economic sectors
- (b) Encourage the citizens to move out for job
- (c) Seek Grant -in Aid from Central Government external agencies
- ☒ (d) Invest adequately on health and education sectors

- (A) Only measure (a) would help
- (B) Measures (a) and (c) would help
- (C) Measures (a), (b) and (c) would help
- (D) All measures would help

49. Consider the statements regarding Raster Data Model

- a. It has a simple data structure
- b. Better suited for the output of graphics
- c. Overlay operations are easily and efficiently represented ✓
- d. Topological relationships are efficiently represented ✓

Options:

- (A) Only (b), (c) and (d) are correct
- (B) Only (a) and (c) are correct
- (C) Only (c) and (d) are correct
- (D) Only (a), (b) and (c) are correct

☒ 50. Which of the following is the recent development in Geographic studies?

- (A) Regional Geography
- (B) Astronomy
- (C) Determination
- ☒ (D) Positivism

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